

# Robust optimal scheduling for integrated energy systems based on multi-objective confidence gap decision theory

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## ABSTRACT

The multiple uncertainties in renewable energy generation and load pose challenges for the operation of integrated energy systems (IES) with robustness. To achieve robust scheduling of IES, this paper proposes a multi-objective bi-level confidence gap decision theory robust model (MOCGDT). First, a neural network-based surrogate model is used to obtain the inverse cumulative distribution function of a Gaussian mixture model for uncertain parameters, constructing a confidence interval uncertainty set. Then, the upper-level optimization objective is to maximize the fluctuations of wind, solar, and load, while the lower-level objective is to minimize the total operating cost, resulting in an IES MOCGDT robust model. Finally, an effective bi-level solution algorithm is developed for the model's characteristics. The case study results demonstrate that the proposed method improves economic costs by 20.69% and 3.87% compared to robust optimization and stochastic optimization, respectively, and reduces computation time by approximately 601 s compared to stochastic optimization. This suggests that the proposed MOCGDT model not only reduces the conservatism of conventional robust decision-making but also overcomes the coarseness of uncertain sets in the information gap decision theory model, achieving more reasonable robust scheduling with uncertainty. The effectiveness and superiority of the proposed method are thus verified.

## 1. Introduction

With the rapid growth of industries and rising energy needs, the efficient and effective utilization and distribution of energy have become a significant research focus (Lawal & Teh, 2023). Integrated energy systems (IES) have gained significant attention, as they offer complementary use of electricity, heating, and cooling energy sources, leading to increased energy efficiency and reduced environmental pollution (Li et al., 2023; Zhang et al., 2023). An IES is a complex energy system with sustainable development, consisting of energy conversion, distribution and storage devices, and various end-users. It can provide electricity, cooling energy, heating energy, and other loads simultaneously. An IES can obtain electrical energy from various distributed sources. In addition, cooling energy can be generated by utilizing adsorption chillers. To manage the energy supply and demand, any excess energy can be stored in electrical or thermal energy storage devices and then utilized during peak demand periods. However, the complexity of coupling between multiple energy sources and the uncertainty associated with renewable energy generation (REG) and load

demand make it challenging to develop an effective dispatching scheme. Therefore, robust scheduling that considers uncertainty is of significant importance to the operation of IES (Mohseni et al., 2022).

Research on the uncertainty optimization scheduling problem has been carried out extensively in the last few decades and can be broadly classified into two main approaches: stochastic optimization (SO) and robust optimization (RO), depending on how uncertainty factors are modeled (Rahmani & Amjady, 2019). In SO, the uncertainty is described using a probability distribution function (PDF), and typical scenarios are generated through sampling. The optimized solution is then the average optimal solution across all scenarios. In (Vahid-Pakdel et al., 2017), under the uncertainty of wind power output and price, an energy hub operation strategy considering demand response and the thermal energy market was proposed based on a SO method. In (Mansouri et al., 2020), the Monte Carlo method is adopted to generate different scenarios, and then the K-means method is used to reduce the scenarios and propose a strategy for optimal planning. In (Zheng et al., 2021), the uncertainty of load and photovoltaic (PV) production in grid-connected residential PV-battery systems is addressed through a proposed two-stage SO method. In (Wang et al., 2022), a multi-stage SO model for an integrated

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**Nomenclature****Abbreviations**

|         |                                                |
|---------|------------------------------------------------|
| IES     | integrated energy system                       |
| CHP     | combined heat and power                        |
| SO      | stochastic optimization                        |
| RO      | robust optimization                            |
| IGDT    | information gap decision theory                |
| MOCGDT  | multi-objective confidence gap decision theory |
| CDF     | cumulative distribution function               |
| GMM     | Gaussian mixture model                         |
| RMSE    | root mean square error                         |
| MAE     | absolute average error                         |
| WT      | wind turbine                                   |
| PV      | photovoltaic                                   |
| EB      | electrical boilers                             |
| EC      | electrical chiller                             |
| AC      | absorption chiller                             |
| ES      | electricity storage                            |
| HS      | heat storage                                   |
| STA     | state transition algorithm                     |
| PMOSTA  | population-based multi-objective STA           |
| NSGA-II | non-dominated sorting genetic algorithm        |
| MOPSO   | multi-objective particle swarm optimization    |
| CNY     | Chinese Yuan                                   |

**Index**

|     |                              |
|-----|------------------------------|
| $t$ | Index of times               |
| $i$ | Index of Gaussian components |

**Parameters**

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| $\lambda$         | objective deviation factor                                                         |
| $c_{e1,t}$        | electricity purchase price in period $t$                                           |
| $c_{e2,t}$        | electricity sale price in period $t$                                               |
| $c_{g,t}$         | natural gas price in period $t$                                                    |
| $c_n^m$           | equipment $n$ maintenance cost                                                     |
| $c_{waste,t}$     | penalty cost coefficient per unit of wind power, PV and cutting load in period $t$ |
| $P_{CHP}^{\min}$  | lower limits of power generation of CHP                                            |
| $P_{CHP}^{\max}$  | upper limits of power generation of CHP                                            |
| $H_{CHP}^{\min}$  | lower limits of the output heat of CHP                                             |
| $H_{CHP}^{\max}$  | upper limits of the output heat of CHP                                             |
| $\gamma_{CHP}$    | thermoelectric ratio of CHP                                                        |
| $\eta_{CHP}$      | power generation efficiency of CHP unit                                            |
| $P_{EB}^{\min}$   | lower limits of power consumed of EB                                               |
| $P_{EB}^{\max}$   | upper limits of power consumed of EB                                               |
| $\gamma_{EB}$     | conversion efficiency of the EB                                                    |
| $H_{AC}^{\min}$   | lower limits of heat consumed by AC                                                |
| $H_{AC}^{\max}$   | upper limits of heat consumed by AC                                                |
| $\gamma_{AC}$     | conversion efficiency of AC                                                        |
| $P_{EC}^{\min}$   | lower limits of power consumed by EC                                               |
| $P_{EC}^{\max}$   | upper limits of power consumed by EC                                               |
| $\gamma_{EC}$     | conversion efficiency of EC                                                        |
| $\eta_{ES}^{ch}$  | the charging efficiencies of ES                                                    |
| $\eta_{ES}^{dis}$ | the discharging efficiencies of ES                                                 |
| $\eta_{HS}^{ch}$  | the charging efficiencies of HS                                                    |

|                      |                                                        |
|----------------------|--------------------------------------------------------|
| $\eta_{HS}^{dis}$    | discharging efficiencies of HS                         |
| $E_{ES}^{\min}$      | lower limits of ES capacity                            |
| $E_{ES}^{\max}$      | upper limits of ES capacity                            |
| $E_{HS}^{\min}$      | lower limits of HS capacity                            |
| $E_{HS}^{\max}$      | upper limits of HS capacity                            |
| $P_{ES}^{ch-\max}$   | upper limits of charging power of ES                   |
| $P_{ES}^{dis-\max}$  | upper limits of discharging power of ES                |
| $H_{HS}^{ch-\max}$   | upper limits of charging heat of HS                    |
| $H_{HS}^{dis-\max}$  | upper limits of discharging heat of HS                 |
| $UR_n$               | ramp-up rate limits of device $n$                      |
| $DR_n$               | ramp-down rate limits of device $n$                    |
| $\tilde{P}_{W,t}$    | predicted values of wind power output                  |
| $\tilde{P}_{S,t}$    | predicted values of photovoltaic output                |
| $\tilde{P}_{L,t}$    | predicted values of load demand                        |
| $C_{L,t}$            | cooling load demand                                    |
| $H_{L,t}$            | heat load demand                                       |
| $\varepsilon_\alpha$ | rotation state transformation factor                   |
| $\varepsilon_\beta$  | translation state transformation factor                |
| $\varepsilon_\gamma$ | expansion state transformation factor                  |
| $\varepsilon_\delta$ | axesion state transformation factor                    |
| $\rho_i$             | weight coefficient of the $i$ th Gaussian distribution |
| $\mu_i$              | expectation of the $i$ th Gaussian distribution        |
| $\sigma_i^2$         | variance of the $i$ th Gaussian distribution           |

**Variables**

|                       |                                             |
|-----------------------|---------------------------------------------|
| $E_t^{HS}$            | capacities of HS in period $t$              |
| $E_t^{ES}$            | capacities of ES in period $t$              |
| $\sigma_{ES,t}^{ch}$  | charging states of ES in period $t$         |
| $\sigma_{ES,t}^{dis}$ | discharging states of ES in period $t$      |
| $\sigma_{HS,t}^{ch}$  | charging heat states of HS in period $t$    |
| $\sigma_{HS,t}^{dis}$ | discharging heat states of HS in period $t$ |
| $P_{ES,t}^{ch}$       | charging power of ES                        |
| $P_{ES,t}^{dis}$      | discharging power of ES                     |
| $H_{HS,t}^{ch}$       | charging heat of HS                         |
| $H_{HS,t}^{dis}$      | discharging heat of HS                      |
| $\Gamma_a$            | robustness targets of wind power            |
| $\Gamma_b$            | robustness targets of photovoltaic power    |
| $\Gamma_c$            | robustness targets of load                  |
| $P_{W,t}$             | wind power output                           |
| $P_{S,t}$             | photovoltaic output                         |
| $P_{L,t}$             | load demand                                 |
| $P_{e1,t}$            | power purchased from power-grid             |
| $P_{e2,t}$            | power sold to the grid                      |
| $P_{CHP,t}$           | output power of CHP unit                    |
| $P_{EC,t}$            | power consumed of EC                        |
| $P_{EB,t}$            | power consumed by EB                        |
| $H_{CHP,t}$           | output heat of the CHP unit                 |
| $H_{EB,t}$            | output heat of EB                           |
| $H_{AC,t}$            | heat consumed by AC                         |
| $C_{EC,t}$            | output cooling of EC                        |
| $C_{AC,t}$            | output cooling of AC                        |
| $\alpha_i$            | significance level                          |

distribution system of resilient electricity and natural gas is proposed. RO involves defining a set of possible values for uncertain parameters and finding a solution that satisfies all constraints for every value in the uncertainty set. The goal of RO is to optimize the objective function such that it performs as well as possible for the worst-case scenario within the

uncertainty set (Peng et al., 2015; Yang & Su, 2021). In (Ebrahimi & Amjady, 2019), an adaptive robust optimization framework for day-ahead microgrid scheduling is proposed, and its effectiveness is verified by test cases. In (Zhong et al., 2021), a three-level RO model for IES is developed to address multiple uncertainties, including the output of

**Table 1**

Comparing this work with recent research for uncertainty optimization.

| Ref                           | Multi-energy |      |         | Uncertainty |    |      | Uncertain modeling methods | Solving algorithm      |
|-------------------------------|--------------|------|---------|-------------|----|------|----------------------------|------------------------|
|                               | Power        | Heat | Cooling | Wind        | PV | Load |                            |                        |
| (Vahid-Pakdel et al., 2017)   | ✓            | ✓    | ×       | ✓           | ×  | ×    | SO                         | CPLEX                  |
| (Mansouri et al., 2020)       | ✓            | ✓    | ✓       | ✓           | ×  | ✓    | SO                         | Benders decomposition  |
| (Zheng et al., 2021)          | ✓            | ×    | ×       | ×           | ✓  | ✓    | SO                         | CPLEX                  |
| (Wang et al., 2022)           | ✓            | ×    | ×       | ×           | ×  | ✓    | SO                         | GUROBI                 |
| (Peng et al., 2015)           | ✓            | ×    | ×       | ✓           | ✓  | ✓    | RO                         | Differential evolution |
| (Ebrahimi & Amjadi, 2019)     | ✓            | ×    | ×       | ✓           | ✓  | ×    | RO                         | Benders decomposition  |
| (Zhong et al., 2021)          | ✓            | ✓    | ×       | ✓           | ✓  | ×    | RO                         | C&CG, ADMM             |
| (Tan et al., 2022)            | ✓            | ✓    | ×       | ✓           | ×  | ×    | RO                         | GUROBI                 |
| (Wu et al., 2022)             | ✓            | ✓    | ✓       | ✓           | ✓  | ✓    | RO                         | C&CG                   |
| (Chen et al., 2022)           | ✓            | ✓    | ✓       | ✓           | ✓  | ✓    | RO                         | C&CG                   |
| (Mehdizadeh et al., 2018)     | ✓            | ×    | ×       | ✓           | ✓  | ✓    | IGDT                       | GAMS                   |
| (Vahid-Ghavidel et al., 2019) | ✓            | ×    | ×       | ×           | ×  | ✓    | IGDT                       | GAMS                   |
| (Liu et al., 2022)            | ✓            | ✓    | ✓       | ×           | ×  | ×    | IGDT                       | –                      |
| (Najafi et al., 2022)         | ✓            | ✓    | ✓       | ✓           | ✓  | ✓    | IGDT-RO                    | CPLEX                  |
| This work                     | ✓            | ✓    | ✓       | ✓           | ✓  | ✓    | MOCGDT                     | PMOSTA, CPLEX          |

distributed generation and the electric vehicle aggregator. In (Tan et al., 2022), a two-stage RO dispatch model is developed for integrated power and heat network. In (Wu et al., 2022), to address heterogeneous uncertainties in the IES planning of the park, a two-stage RO model is proposed to obtain the equipment configuration scheme of each planning stage. In (Chen et al., 2022), an adjustable two-stage robust energy management optimization framework is established considering the REG output and load uncertainties. It should be noted that both SO and RO have shortcomings: too many scene trees in SO will lead to a sudden increase in the amount of calculation, and scene reduction not only loses the ergodicity of the uncertain interval but also cannot guarantee robustness. If the clustering algorithm is not accurate enough, the typical scene after the scene reduction is not representative enough. RO is generally based on the preset robustness coefficient for robustness analysis, and its robustness setting is often too subjective or conservative.

Information gap decision theory (IGDT) is an efficient model for managing uncertain risks that involves analyzing the potential impact of uncertain parameters while still meeting predetermined objectives (Izadi et al., 2023). In (Mehdizadeh et al., 2018), risk-based energy management of renewable energy-based microgrids in the presence of peak load management is performed using information gap decision theory. In (Vahid-Ghavidel et al., 2019), an IGDT-based robust decision model for short-term self-scheduling of demand response (DR) aggregators is proposed, considering the impact of uncertainty in market electricity prices and consumer participation on the scheduling strategy. In (Liu et al., 2022), a multi-energy retailer scheduling strategy with DR based on IGDT is proposed. In (Najafi et al., 2022), a novel optimization framework is proposed for optimal self-scheduling of an energy hub, which utilizes a combination of IGDT and RO. However, The IGDT method is inadequate in dealing with uncertain variables as it does not consider their distribution probability characteristics. Moreover, most IGDT models only consider the impact of a single uncertain variable and fail to fully analyze the relationship between multiple uncertain variables. This study aims to optimize the dispatch of IES under multiple uncertainties, including wind, PV, and load variations, with the objective of minimizing economic costs. Given the limitations of existing optimization models such as RO, SO, and the basic IGDT model for handling uncertain optimization problems, a data-driven multi-objective confidence gap decision theory (MOCGDT) model is proposed to solve the IES dispatch problem. Table 1 intuitively summarizes the comprehensive comparison between this work and existing research work in solving energy management with uncertainty. The main contributions of this study can be summarized as follows:

- (1) To overcome the limitations of the basic IGDT model, which uses a symmetric box set to describe uncertain parameters, a Gaussian

mixture model (GMM) is used to fit the probability distribution of the prediction error of the uncertain parameters. An surrogate-based model is then used to solve the inverse cumulative distribution function, which provides confidence intervals for the uncertain parameters. The use of confidence intervals instead of box sets to describe the range of uncertain parameters improves the objectivity and accuracy of the model.

- (2) To address the issue of IGDT's inability to handle multiple uncertain parameters simultaneously, the multi-objective optimization theory is incorporated into the IGDT model, along with the confidence interval modeling of uncertain parameters, resulting in the MOCGDT robust model for solving multiple uncertain optimization problems.
- (3) To validate the effectiveness of the proposed MOCGDT robust model, it is applied to IES scheduling problems with wind, PV, and load uncertainties. Since the MOCGDT is a bi-level model, an efficient bi-level solution algorithm is designed based on the characteristics of the model. A multi-objective intelligent optimization algorithm is used to solve the upper-level multi-objective problem, and the CPLEX solver is used to solve the lower-level mixed-integer programming problem.

The rest of this article will be presented as follows. In Section 2, the proposed MOCGDT is described in detail. In Section 3, the MOCGDT-based robust model of IES is proposed and the solving algorithm for the model is designed. In Section 4, the effectiveness of the proposed method is verified on a specific IES case. Finally, the conclusion is given in Section 5.

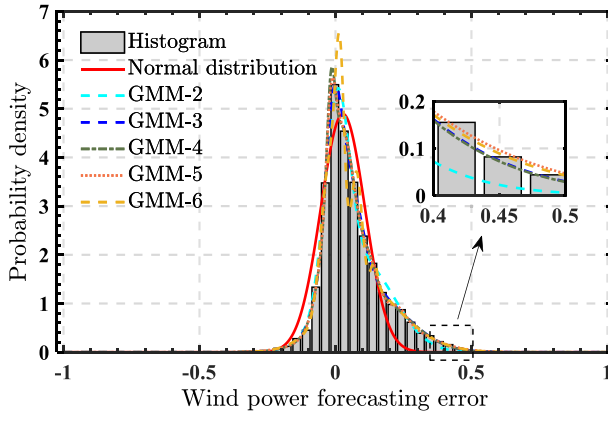
## 2. Proposed method

In this section, we introduce the construction process of the MOCGDT in detail. Firstly, the confidence interval of uncertain variables is constructed based on GMM in a data-driven way. Then the confidence interval is used to replace the box uncertainty set in IGDT. Finally, combined with multi-objective optimization technology, the MOCGDT model is proposed.

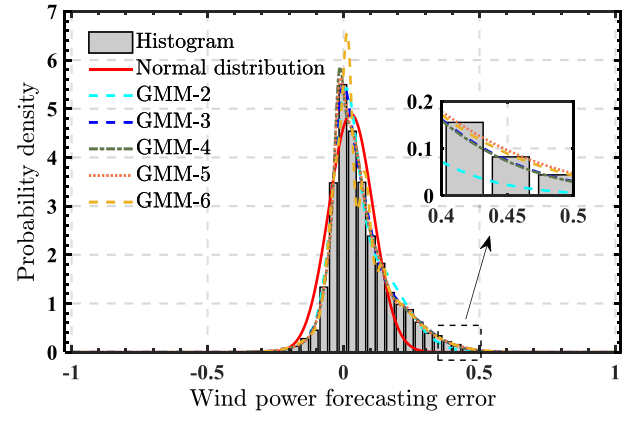
### 2.1. Uncertainty variable description based on data-driven

#### 2.1.1. Fitting probability distribution by GMM

In real-world scenarios where data is collected from multiple sources with varying characteristics, the presence of complex and unknown noise is inevitable. In such cases, it is prudent to model the probability distribution of the real noise, especially when its true distribution remains elusive. This approach can help improve the accuracy and reliability of the data analysis and enable more robust predictions and



(a) PDF curve of wind power forecasting error



(b) PDF fitting curve of PV forecasting error

Fig. 1. PDF curves of wind power and PV forecasting error with various GMM.

Table 2

Accuracy comparison of fitting error distribution of wind power forecasting with GMM.

| Index | Normal distribution | GMM-2  | GMM-3  | GMM-4  | GMM-5  | GMM-6  |
|-------|---------------------|--------|--------|--------|--------|--------|
| MAE   | 0.1520              | 0.0453 | 0.0152 | 0.0092 | 0.0064 | 0.0063 |
| RMSE  | 0.3103              | 0.0949 | 0.0292 | 0.0189 | 0.0139 | 0.0141 |

Table 3

Accuracy comparison of fitting error distribution of PV forecasting with GMM.

| Index | Normal distribution | GMM-2  | GMM-3  | GMM-4  | GMM-5  | GMM-6  |
|-------|---------------------|--------|--------|--------|--------|--------|
| MAE   | 0.1139              | 0.0447 | 0.0427 | 0.0348 | 0.0304 | 0.0250 |
| RMSE  | 0.1667              | 0.0628 | 0.0613 | 0.0539 | 0.0424 | 0.0456 |

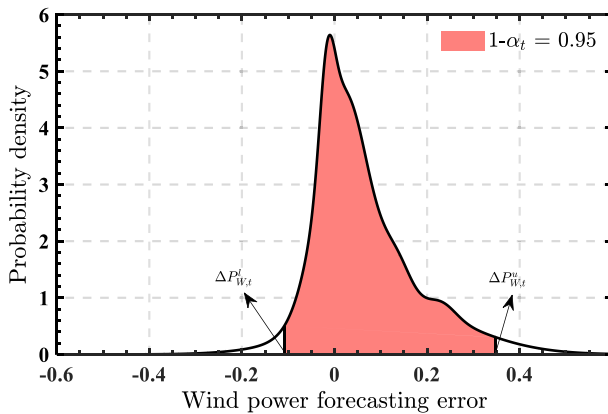
insights (Wang, Wang, Hu, 2017). GMM has been proven to have better approximation capability for the continuous distribution of complex noise (Dong et al., 2021; Zhao et al., 1995). Therefore, GMM is employed to fit the wind-PV-load forecasting error distribution in a data-driven way. The fluctuation intervals of the wind-PV-load uncertain variables are constructed at a given confidence level, thus overcoming the shortcomings of the boxed uncertainty set. Assuming that the unknown

errors are random variables, the mathematical formula of the GMM can be observed in Eq. (1).

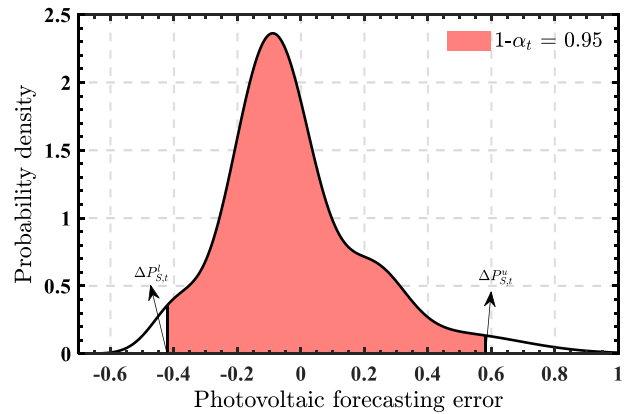
$$\begin{cases} f_{GMM} = \sum_{i=1}^k \rho_i N_i(u | \mu_i, \sigma_i^2) \\ \text{s.t.} \begin{cases} 0 \leq \rho_i \leq 1 \\ \sum_{i=1}^k \rho_i = 1 \end{cases} \end{cases} \quad (1)$$

where  $k$  is the number of Gaussian components;  $N_i(u | \mu_i, \sigma_i^2)$  represents the  $i$ th Gaussian distribution;  $\rho_i, \mu_i, \sigma_i^2$  are the weight coefficient, expectation, and variance of the  $i$ th Gaussian distribution, respectively.

This paper utilizes the historical wind power and PV data of a particular location from January 2021 to October 2021 as a case study. The wind-PV forecasting error's probability distribution for this period is fitted using GMM with 2–6 Gaussian components and is compared with the normal distribution. Note that most of the current studies use the normal distribution to deal with the load uncertainty due to the relatively weak load fluctuations. Therefore, for simplicity, this paper uses a normal distribution to describe the load forecasting error. Fig. 1 shows the PDF of wind power and PV forecasting error. To quantify the fitting effect of each model, Tables 2 and 3 show the fitting metrics of GMM and normal distribution including root mean square error (RMSE) and mean absolute error (MAE). Eqs. (2) and (3) provide the formulas

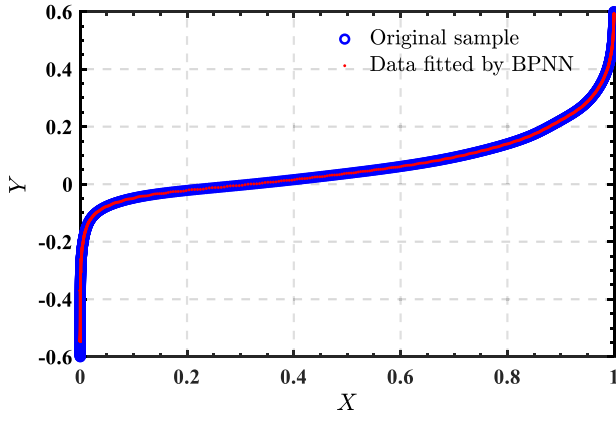


(a) Probability confidence intervals of wind power forecasting error

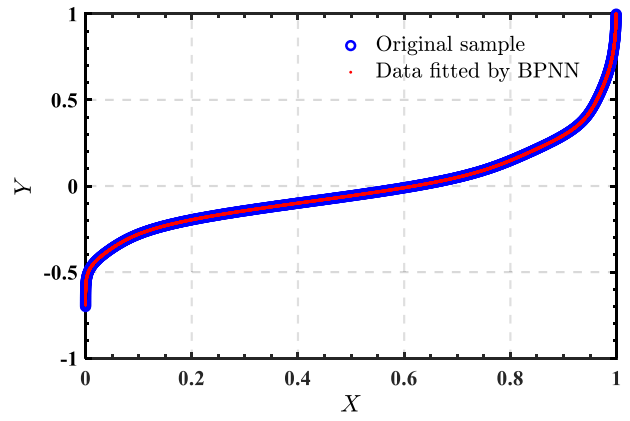


(b) Probability confidence intervals of PV forecasting error

Fig. 2. Probability confidence intervals of wind power and PV forecasting error.



(a) Inverse CDF obtained by BPNN for wind power forecasting error



(b) Inverse CDF obtained by BPNN forecasting error

Fig. 3. The inverse CDF is obtained by the surrogate-based model.

for MAE and RMSE, respectively.

$$MAE = \frac{1}{N_s} \sum_{t=1}^{N_s} |y_t - \hat{y}_t| \quad (2)$$

$$RMSE = \sqrt{\frac{1}{N_s} \sum_{t=1}^{N_s} (y_t - \hat{y}_t)^2} \quad (3)$$

where  $y_t$  is the actual value; and  $\hat{y}_t$  is the value fitted to the probability distribution model;  $N_s$  is the number of samples.

From the fitting results in Tables 2 and 3, the fitting results of GMM are better than that of normal distribution, and with the increase of the number of Gaussian components of GMM, the MAE and RMSE obtained by GMM are smaller, indicating that the fitting effect is getting better and better. However, there is little difference in fitting accuracy between GMM-4, GMM-5, and GMM-6. Therefore, GMM-4 is selected in this paper to fit the probability distribution of forecasting error by making a trade-off between computational efficiency and fitting accuracy.

### 2.1.2. Inverse CDF

From Section 2.1.1, it is easy to understand that after obtaining the PDF of wind-PV-load forecasting error, the upper and lower limits of the forecasting error can be calculated by setting the value of the confidence level  $1 - \alpha_t$  of the probability distribution at different moments and thus constructing the fluctuation interval of the uncertain variables. As shown in Fig. 2, when  $1 - \alpha_t = 0.9$  is set, the forecasting errors of wind power and PV can be obtained by fitting them with GMM-4. In Fig. 2,  $\Delta P_{W,t}^l$  and  $\Delta P_{W,t}^u$  are the maximum values of the negative and positive deviations of the wind power forecasting error;  $\Delta P_{S,t}^l$  and  $\Delta P_{S,t}^u$  are the maximum values of the negative and positive deviations of the PV forecasting error, respectively. Obviously, to construct the fluctuation intervals for wind power and PV at a given confidence level  $1 - \alpha_t$ , it is necessary to solve for the specific values of  $\Delta P_{W,t}^l$ ,  $\Delta P_{W,t}^u$ ,  $\Delta P_{S,t}^l$  and  $\Delta P_{S,t}^u$  through the inverse cumulative distribution function (CDF). Take wind power as an example, as shown in Eq. (4).

$$\Delta P_{W,t}^l = F_{W,t}^{-1}\left(\frac{\alpha_t}{2}\right), \Delta P_{W,t}^u = F_{W,t}^{-1}\left(1 - \frac{\alpha_t}{2}\right) \quad (4)$$

where  $F_{W,t}^{-1}(\cdot)$  is the inverse CDF of wind power forecasting error.

Due to the complexity of the mathematical expressions of the GMM, it is difficult to solve its inverse CDF directly. Some scholars fit the CDF of GMM by high-order polynomials and then obtain the solution of its inverse function by the polynomial root method (Wang, Shen et al.,

2017). However, when the order of the polynomial is greater than 4, it is difficult to solve the analytic roots of the polynomial (Klein, 2003; Serre, 1979). Given this defect, back propagation neural network (BPNN) (Wang et al., 2011) with good fitting characteristics is used as the surrogate model to directly fit the inverse CDF, and a method to solve the inverse CDF surrogate-based model is proposed to avoid the complex process of indirectly fitting the CDF and then solving the root of the inverse CDF. The reason why the surrogate model can accurately describe the inverse CDF of GMM is that the following three conditions are met simultaneously:

- (1) The CDF is monotonic, and its inverse CDF is also monotonic, so there is a one-to-one mapping relationship between the input and output of the constructed proxy model.
- (2) The selected sample points are evenly distributed in the whole variable space and the sample size is large enough.
- (3) The surrogate model has a strong fitting ability.

Since Condition (1) is an inherent property of the CDF, the above three conditions can be satisfied only by selecting appropriate samples and the surrogate model, and the surrogate model can well fit the characteristics of the inverse CDF. Specifically, a large number of sample points  $(X, Y)$  are obtained by integrating the PDF. Note that  $X$  represents the probability value and  $Y$  represents the random variable. Then;  $X$  is taken as the input and  $Y$  as the output, BPNN is used to fit the sample, to obtain the inverse CDF. Fig. 3 shows the inverse CDF obtained by the BPNN surrogate model when the number of Gaussian components  $k$  is 4.

### 2.1.3. Confidence interval modeling of wind-PV-load uncertainty

After obtaining the inverse CDF, confidence intervals for the wind-PV-load uncertainty variables in the IES can be constructed for each period based on the given confidence level, as described below.

$$\begin{cases} P_{W,t} \in U_i(\Delta P_{W,t}, \alpha_{W,t}) = \left\{ P_{W,t} \mid \tilde{P}_{W,t} + \Delta P_{W,t}^l \leq P_{W,t} \leq \tilde{P}_{W,t} + \Delta P_{W,t}^u \right\} \\ \Delta P_{W,t} \in \left\{ \Delta P_{W,t}^l, \Delta P_{W,t}^u \right\} \\ \Delta P_{W,t}^l = F_{W,t}^{-1}\left(\frac{\alpha_{W,t}}{2}\right), \Delta P_{W,t}^u = F_{W,t}^{-1}\left(1 - \frac{\alpha_{W,t}}{2}\right) \end{cases} \quad (5)$$



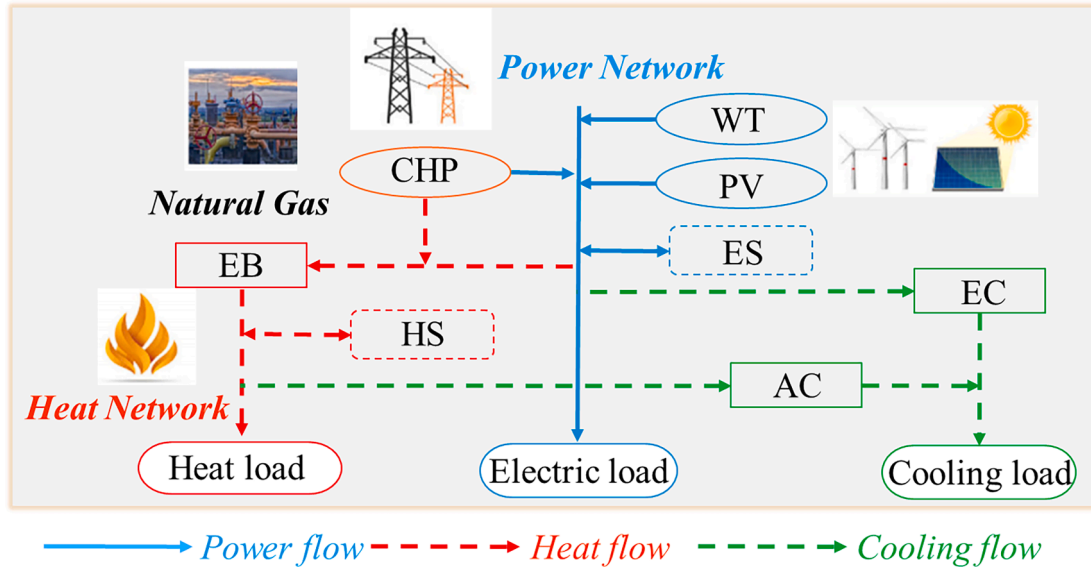


Fig. 4. Structure diagram of IES.

$$\begin{cases} P_{S,t} \in U_i(\Delta P_{S,t}, \alpha_{S,t}) = \left\{ P_{S,t} \mid \tilde{P}_{S,t} + \Delta P_{S,t}^l \leq P_{S,t} \leq \tilde{P}_{S,t} + \Delta P_{S,t}^u \right\} \\ \Delta P_{S,t} \in \left\{ \Delta P_{S,t}^l \mid \Delta P_{S,t}^l \leq \Delta P_{S,t} \leq \Delta P_{S,t}^u \right\} \\ \Delta P_{S,t}^l = F_{S,t}^{-1}\left(\frac{\alpha_{S,t}}{2}\right), \Delta P_{S,t}^u = F_{S,t}^{-1}\left(1 - \frac{\alpha_{S,t}}{2}\right) \end{cases} \quad (6)$$

$$\begin{cases} P_{L,t} \in U_i(\Delta P_{L,t}, \alpha_{L,t}) = \left\{ P_{L,t} \mid \tilde{P}_{L,t} + \Delta P_{L,t}^l \leq P_{L,t} \leq \tilde{P}_{L,t} + \Delta P_{L,t}^u \right\} \\ \Delta P_{L,t} \in \left\{ \Delta P_{L,t}^l \mid \Delta P_{L,t}^l \leq \Delta P_{L,t} \leq \Delta P_{L,t}^u \right\} \\ \Delta P_{L,t}^l = F_{L,t}^{-1}\left(\frac{\alpha_{L,t}}{2}\right), \Delta P_{L,t}^u = F_{L,t}^{-1}\left(1 - \frac{\alpha_{L,t}}{2}\right) \end{cases} \quad (7)$$

where  $P_{W,t}$ ,  $P_{S,t}$ , and  $P_{L,t}$  are the actual observation values of wind power, PV, and load in period  $t$ , respectively;  $\tilde{P}_{W,t}$ ,  $\tilde{P}_{S,t}$ , and  $\tilde{P}_{L,t}$  are respectively the predicted values in period  $t$ ;  $\alpha_{W,t}$ ,  $\alpha_{S,t}$ , and  $\alpha_{L,t}$  are their significance levels;  $\Delta P_{W,t}^l$ ,  $\Delta P_{S,t}^l$ , and  $\Delta P_{L,t}^l$  are respectively the lower limit of the confidence interval of wind power, PV, and load forecasting error in period  $t$ ;  $\Delta P_{W,t}^u$ ,  $\Delta P_{S,t}^u$ , and  $\Delta P_{L,t}^u$  are the upper limits of the confidence interval in period  $t$ , respectively;

## 2.2. Proposed multi-objective confidence gap decision theory

The IGDT robust model (Ben-Haim, 2006) is designed to ensure that the impact of uncertainty on optimization results is maximized, while still guaranteeing that the optimization result will not be worse than a predetermined result (Tostado-Véliz et al., 2023). The mathematical model is as follows.

$$\begin{cases} \max_X \\ \text{s.t.} \end{cases} \begin{cases} f(X, u) \leq f_c, f_c = (1 + \lambda)f_0 \\ h(X, u) = 0 \\ g(X, u) \leq 0 \\ u \in U(\tilde{u}, \varepsilon) = \left\{ u \mid \left| \frac{u - \tilde{u}}{\tilde{u}} \right| \leq \varepsilon \right\} \end{cases} \quad (8)$$

where  $X$  is the variable to be optimized;  $u$  is the uncertain parameter of the optimization model;  $U(\tilde{u}, \varepsilon)$  denotes the fluctuation interval of the

uncertain parameter, here,  $\tilde{u}$  and  $\varepsilon$  are predicted values and fluctuation bias factors of uncertain parameters, respectively;  $f_0$  is the optimal objective function value obtained when  $u$  is equal to  $\tilde{u}$  under the deterministic optimization model;  $\lambda$  is the objective deviation factor, the larger its value is set, the more tolerant the model is to objective variation and the more robust it is, and vice versa.  $f_c$  is the worst objective value acceptable to the decision-maker after accounting for uncertainty.

It is not difficult to understand the robustness of the aforementioned IGDT model when dealing with uncertain optimization problems. However, it should be emphasized that the basic IGDT model can only handle a single uncertain optimization problem. In addition, the box uniform symmetry fluctuation interval  $U(\tilde{u}, \varepsilon)$  is too rough to describe the uncertainty, does not fully consider the distribution probability characteristics of uncertain parameters, and cannot accurately reflect the fluctuation interval of uncertain variables in practice. Given the above shortcomings, in this paper, the confidence interval of the probability distribution is employed instead of the boxed uncertainty set in the IGDT model, and the deviation coefficient of uncertainty quantity in the IGDT is eliminated. The proposed MOCGDT model combines multi-objective optimization with IGDT to handle multiple uncertainties. Through the MOCGDT model, decision-makers can analyze and select the most optimal solution from a set of options, considering multiple objectives and uncertainties. In essence, the MOCGDT model provides a robust and comprehensive framework for decision-making that can effectively manage multiple uncertainties.

$$\begin{cases} \max_X \{1 - \alpha_1, 1 - \alpha_2, 1 - \alpha_3, \dots, 1 - \alpha_m\} \\ \text{s.t.} \end{cases} \begin{cases} 1 - \alpha_i = \Pr_i\{\Delta u_i^l \leq \Delta u_i \leq \Delta u_i^u\} \\ 0 \leq \alpha_i \leq 1, i = 1, 2, 3, \dots, m \\ f(X, u_1, u_2, \dots, u_n) \leq f_c, f_c = (1 + \lambda)f_0 \\ h(X, u_1, u_2, \dots, u_n) = 0 \\ g(X, u_1, u_2, \dots, u_n) \leq 0 \\ u_i \in U_i(\Delta u_i, \alpha_i) = \{u_i \mid \tilde{u}_i + \Delta u_i^l \leq u_i \leq \tilde{u}_i + \Delta u_i^u\} \\ \Delta u_i \in \{\Delta u_i \mid \Delta u_i^l \leq \Delta u_i \leq \Delta u_i^u\} \\ \Delta u_i^l = F_i^{-1}\left(\frac{\alpha_i}{2}\right), \Delta u_i^u = F_i^{-1}\left(1 - \frac{\alpha_i}{2}\right) \end{cases} \quad (9)$$

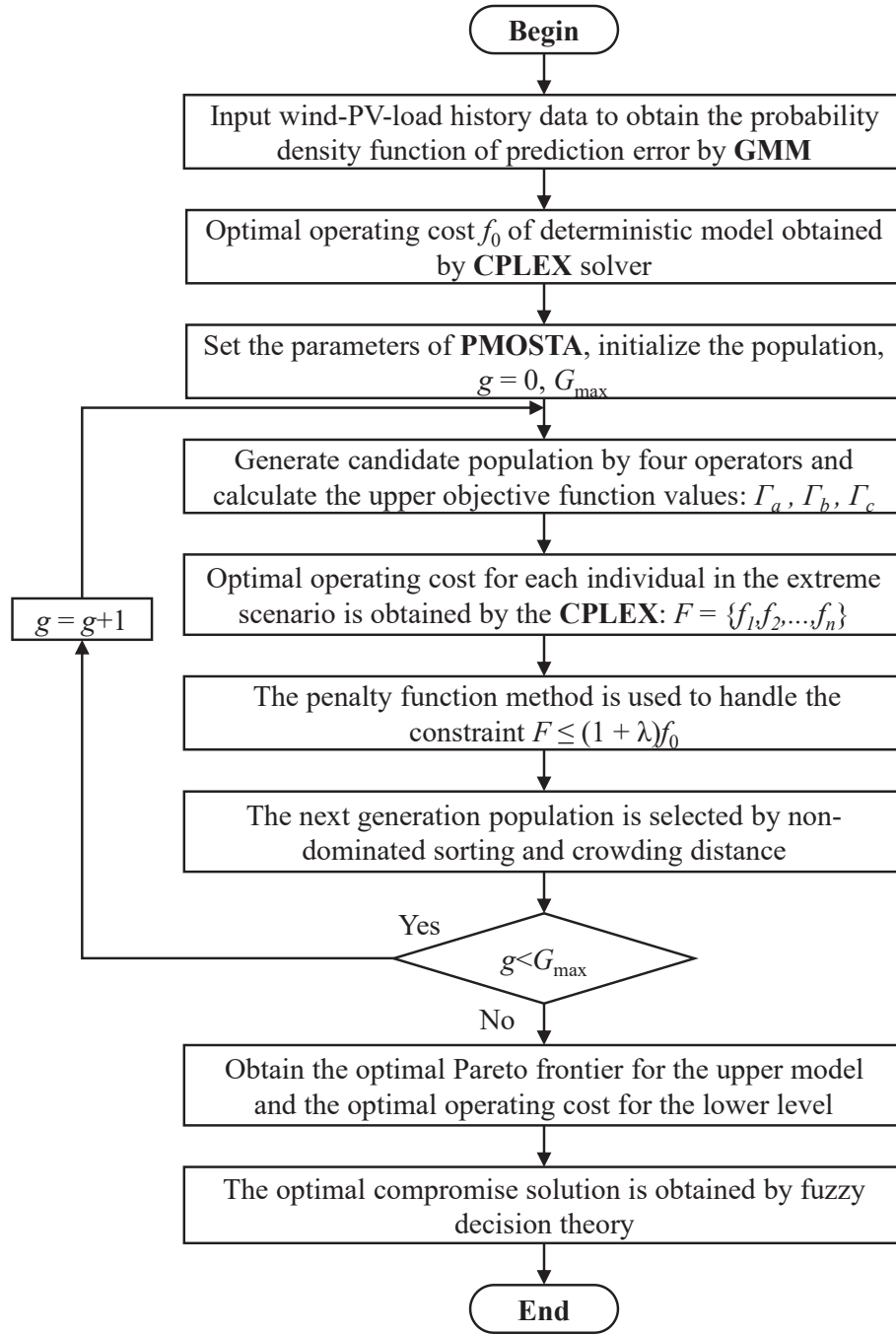


Fig. 5. Framework diagram of the proposed method for solving IES robust optimal scheduling.

where  $m$  is the number of uncertain parameters and  $u_i$  is the  $i$ th uncertainty;  $\Pr_i(\cdot)$ ,  $1 - \alpha_i$  and  $F_i^{-1}(\cdot)$  denotes the probability distribution, confidence level, and inverse CDF of the deviation of the  $i$ th uncertain parameter, respectively;  $\tilde{u}_i$ ,  $\Delta u_i$ ,  $\Delta u_i^l$  and  $\Delta u_i^u$  is the predicted value, deviation quantity, negative deviation and positive deviation, respectively. Generally,  $\Delta u_i^l < 0$ ,  $\Delta u_i^u > 0$ .

Obviously, the proposed MOCGDT model can handle multiple uncertainty optimization problems simultaneously compared to the basic IGDT model. In addition, the uncertainty confidence intervals based on probability distributions can more accurately reflect the fluctuation intervals of uncertain variables in practice. The optimization process is achieved by driving the confidence level of each uncertain variable to be maximized, thus maximizing the ability to avoid the influence of uncertain variables on the optimization results.

### 3. MOCGDT-based robust scheduling for IES

#### 3.1. Mathematical formulation of IES

The IES studied in this paper mainly includes electrical chiller (EC), absorption chiller (AC), wind turbine (WT), PV, combined heat and power (CHP), electrical boilers (EB), electricity storage (ES), heat storage (HS), power load, heating load, and cooling load. Its structure is shown in Fig. 4. This paper constructs a robust scheduling model aiming at minimizing the operation cost.

##### 3.1.1. Objective function

The objective function of the optimization is the total operational cost of the system, as shown in Eq. (10).

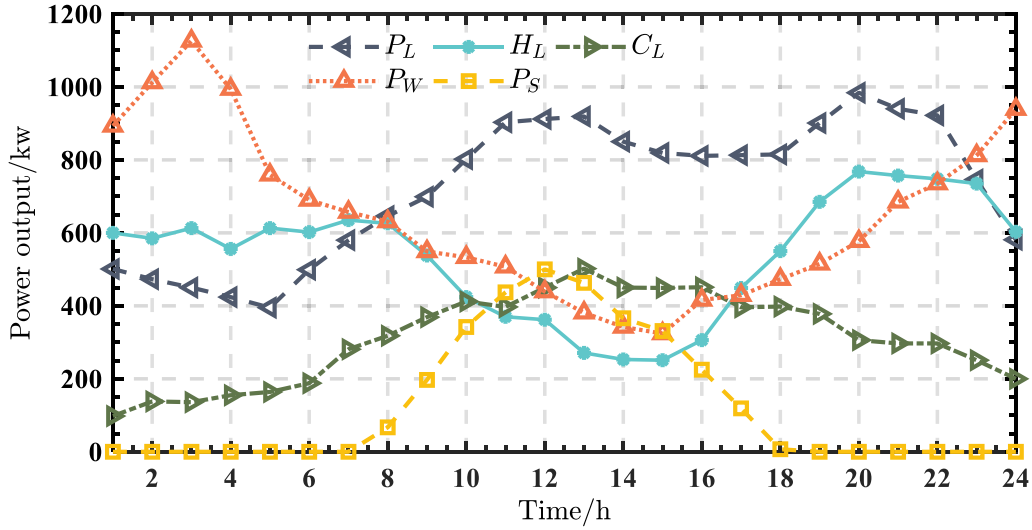


Fig. 6. Forecasted values of wind power, PV and loads.

Table 4

Parameters of IES equipment.

| Parameter                             | Value       | Unit   |
|---------------------------------------|-------------|--------|
| $P_{CHP}^{\min}, P_{CHP}^{\max}$      | 50, 550     | kw     |
| $\eta_{CHP}, \gamma_{CHP}$            | 0.35, 0.8   | –      |
| $UR_{CHP}, DR_{CHP}$                  | 5, 6        | kw/min |
| $c_{CHP}^{om}$                        | 0.1685      | CNY/kw |
| $P_{EB}^{\min}, P_{EB}^{\max}$        | 20, 500     | kw     |
| $\gamma_{EB}$                         | 0.85        | –      |
| $UR_{EB}, DR_{EB}$                    | 3.5         | kw/min |
| $c_{EB}^{om}$                         | 0.0853      | CNY/kw |
| $C_{EC}^{\min}, C_{EC}^{\max}$        | 0, 400      | kw     |
| $\gamma_{EC}$                         | 0.8         | –      |
| $UR_{EC}, DR_{EC}$                    | 4.4         | kw/min |
| $c_{EC}^{om}$                         | 0.016       | CNY/kw |
| $C_{AC}^{\min}, C_{AC}^{\max}$        | 0, 400      | kw     |
| $\gamma_{AC}$                         | 3           | –      |
| $UR_{AC}, DR_{AC}$                    | 5.5         | kw/min |
| $c_{AC}^{om}$                         | 0.0104      | CNY/kw |
| $E_{ES}^{\min}, E_{ES}^{\max}$        | 320, 1000   | kw-h   |
| $\eta_{ES}^{dis}, \eta_{ES}^{ch}$     | 0.95, 0.95  | –      |
| $P_{ES}^{ch-\max}, P_{ES}^{dis-\max}$ | 150, 150    | kw     |
| $E_{HS}^{\min}, E_{HS}^{\max}$        | 100, 600    | kw-h   |
| $\eta_{HS}^{ch}, \eta_{HS}^{dis}$     | 0.95, 0.95  | –      |
| $H_{HS}^{ch-\max}, H_{HS}^{dis-\max}$ | 90, 90      | kw     |
| $c_{wind}^{om}, c_{PV}^{om}$          | 0.04, 0.027 | CNY/kw |

Table 5

Electricity purchase price and sale price.

| Time period                          | Electricity prices (CNY/kw-h) | Electricity selling price (CNY/kw-h) |
|--------------------------------------|-------------------------------|--------------------------------------|
| 0:00–7:00                            | 0.17                          | 0.13                                 |
| 7:00–11:00, 15:00–20:00, 21:00–24:00 | 0.49                          | 0.38                                 |
| 11:00 15:00, 20 21:00                | 0.83                          | 0.65                                 |

## 3.1.2. The constraints of IES

The energy balance constraints for electrical power, heating energy, and cooling energy in the IES are presented in Eqs. (11)–(13).

$$P_{e1,t} + P_{W,t} + P_{S,t} + P_{CHP,t} + P_{ES,t}^{dis} = P_{L,t} + P_{EC,t} + P_{EB,t} + P_{e2,t} + P_{ES,t}^{ch} \quad (11)$$

$$H_{CHP,t} + H_{EB,t} + H_{HS,t}^{dis} = H_{L,t} + H_{AC,t} + H_{HS,t}^{ch} \quad (12)$$

$$C_{EC,t} + C_{AC,t} = C_{L,t} \quad (13)$$

where  $P_{ES,t}^{ch}/H_{HS,t}^{ch}$  and  $P_{ES,t}^{dis}/H_{HS,t}^{dis}$  are the charging and discharging of ES/HS, respectively;  $P_{CHP,t}$ ,  $P_{EC,t}$  and  $P_{EB,t}$  represent the power output of CHP, the power consumed by EC, and the power consumed by EB, respectively;  $H_{CHP,t}$ ,  $H_{EB,t}$ ,  $H_{AC,t}$  and  $H_{L,t}$  the output heat of the CHP, the output heat of EB, the heat consumed by AC and heat load in period  $t$ , respectively;  $C_{EC,t}$ ,  $C_{AC,t}$  and  $C_{L,t}$  represent the cooling output of EC, the cooling output of AC, and the cooling load, respectively.

The constraints of CHP unit:

$$P_{CHP}^{\min} \leq P_{CHP,t} \leq P_{CHP}^{\max} \quad (14)$$

$$H_{CHP}^{\min} \leq H_{CHP,t} \leq H_{CHP}^{\max} \quad (15)$$

$$P_{CHP,t} = \eta_{CHP} P_{g,t} \quad (16)$$

$$P_{CHP,t} = \eta_{CHP} P_{g,t} \quad (17)$$

where  $P_{CHP}^{\min}$  and  $P_{CHP}^{\max}$  represent the CHP unit's minimum and maximum power generation limits, while  $H_{CHP}^{\min}$  and  $H_{CHP}^{\max}$  represent its minimum and maximum output heat limits.  $\eta_{CHP}$  and  $\gamma_{CHP}$  denote the power generation efficiency, and the thermoelectric ratio of CHP unit, respectively.

The constraints of EB:

$$P_{EB}^{\min} \leq P_{EB,t} \leq P_{EB}^{\max} \quad (18)$$

$$f = \min \left( \sum_{t=1}^T (c_{e1,t} P_{e1,t} - c_{e2,t} P_{e2,t}) + \sum_{t=1}^T c_{g,t} P_{g,t} + \sum_{n=1}^N \sum_{t=1}^T c_n^{om} P_{n,t} + \sum_{t=1}^T c_{waste,t} P_{waste,t} \right) \quad (10)$$

where  $T$  is the scheduling period;  $N$  indicates the total number of system equipment;  $c_{e1,t}$ ,  $c_{e2,t}$ ,  $c_{g,t}$  and  $c_n^{om}$  are the purchase price of electricity, electricity sale price, natural gas price, and equipment  $n$  maintenance cost of IES in period  $t$ , respectively;  $P_{e1,t}$ ,  $P_{e2,t}$ ,  $P_{g,t}$  and  $P_{n,t}$  are the purchased power, sold power, purchased gas and equipment output in the period  $t$ , respectively.  $c_{waste,t}$  represents the penalty cost coefficient per unit of wind curtailment, solar curtailment, and cutting load, while  $P_{waste,t}$  denotes the amount of wind curtailment, solar curtailment, and cutting load during the period  $t$ . Note: the tie line of IES has only one status of power purchase or power sale during period  $t$ .



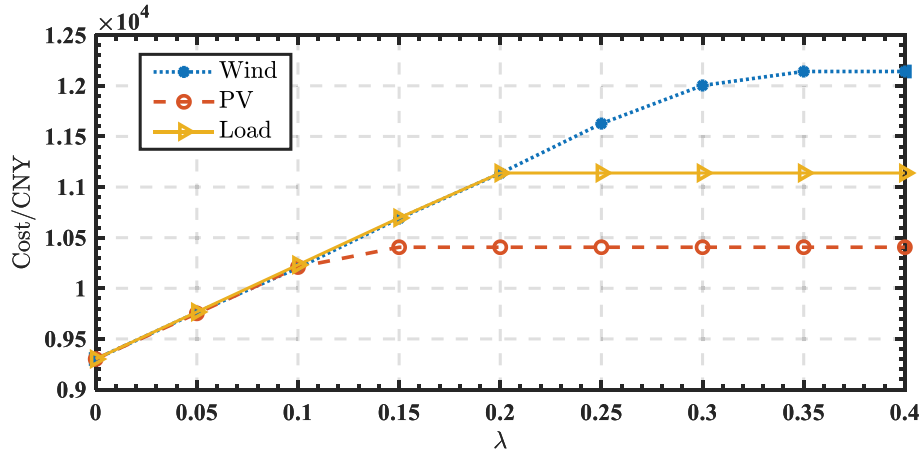


Fig. 7. Economic cost for various λ values.

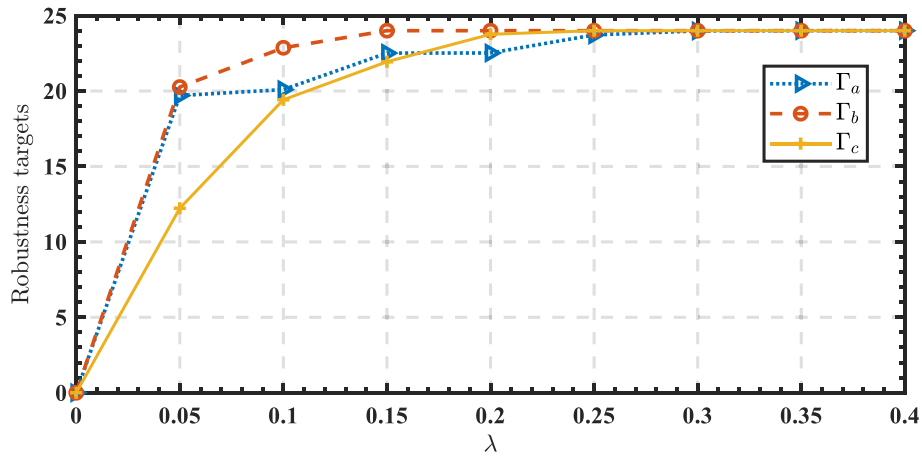


Fig. 8. Robustness targets for various λ values.

$$H_{EB,t} = \gamma_{EB} P_{EB,t} \quad (19)$$

where  $P_{EB}^{\min}$  and  $P_{EB}^{\max}$  lower and upper bounds of the power consumption limits for EB, respectively, while  $\gamma_{EB}$  denotes its conversion efficiency.

The constraints of AC:

$$H_{AC}^{\min} \leq H_{AC,t} \leq H_{AC}^{\max} \quad (20)$$

$$C_{AC,t} = \gamma_{AC} H_{AC,t} \quad (21)$$

where  $H_{AC}^{\min}$  and  $H_{AC}^{\max}$  represent the minimum and maximum heat consumption limits of AC, respectively, while  $\gamma_{AC}$  denotes its conversion efficiency.

The constraints of EC:

$$P_{EC}^{\min} \leq P_{EC,t} \leq P_{EC}^{\max} \quad (22)$$

$$C_{EC,t} = \gamma_{EC} P_{EC,t} \quad (23)$$

where  $P_{EC}^{\min}$  and  $P_{EC}^{\max}$  represent the minimum and maximum power consumption limits of EC, respectively, while  $\gamma_{EC}$  denotes its conversion efficiency.

The constraints of ES/HS: In IES, the state of charge (SOC) of the ES/HS device varies as in Eqs. (24) and (25). In addition, to prevent the energy storage device from overcharging and over discharging, it is necessary to constrain its SOC variability range.

$$E_t^{ES} = E_{t-1}^{ES} + \Delta t \left( P_{ES,t}^{ch} \eta_{ES}^{ch} - \frac{P_{ES,t}^{dis}}{\eta_{ES}^{dis}} \right) \quad (24)$$

$$E_t^{HS} = E_{t-1}^{HS} + \Delta t \left( H_{HS,t}^{ch} \eta_{HS}^{ch} - \frac{H_{HS,t}^{dis}}{\eta_{HS}^{dis}} \right) \quad (25)$$

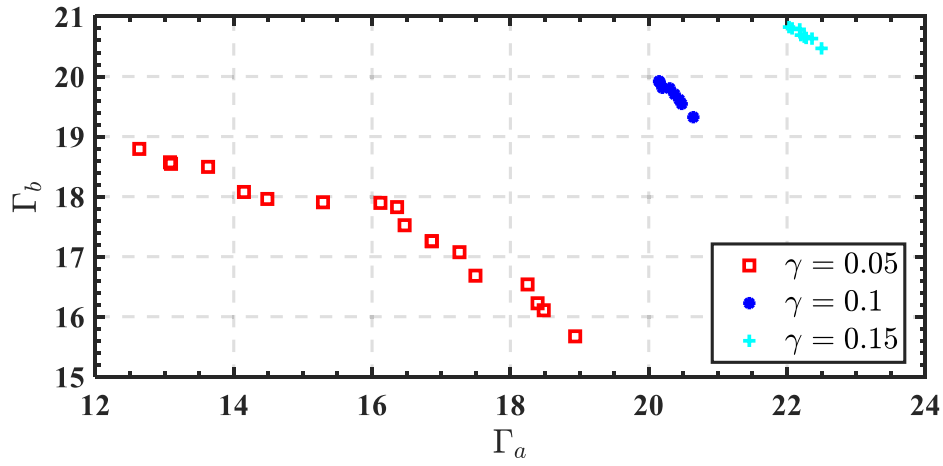
$$E^{ES-\min} \leq E_t^{ES} \leq E^{ES-\max} \quad (26)$$

$$E^{HS-\min} \leq E_t^{HS} \leq E^{HS-\max} \quad (27)$$

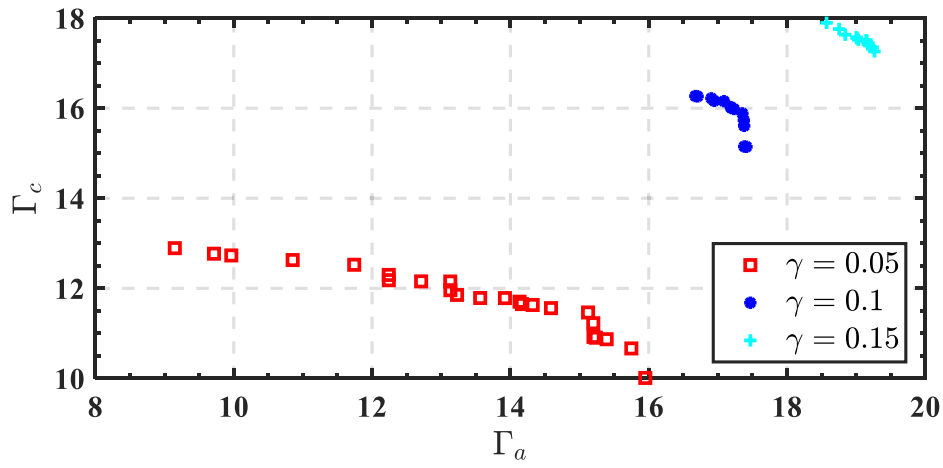
where  $E_t^{ES}/E_t^{HS}$  are the capacities of ES/HS in period  $t$ ;  $\eta_{ES}^{ch}/\eta_{HS}^{ch}$  and  $\eta_{ES}^{dis}/\eta_{HS}^{dis}$  denote efficiencies of ES/HS, respectively.  $E^{ES-\min}/E^{HS-\min}$  and  $E^{ES-\max}/E^{HS-\max}$  represent the lower and upper bounds of the capacity limits for ES/HS, respectively.

Furthermore, the energy storage device's charging and discharging operations cannot occur simultaneously, and their respective powers are limited as specified in Eqs. (28)-(29).

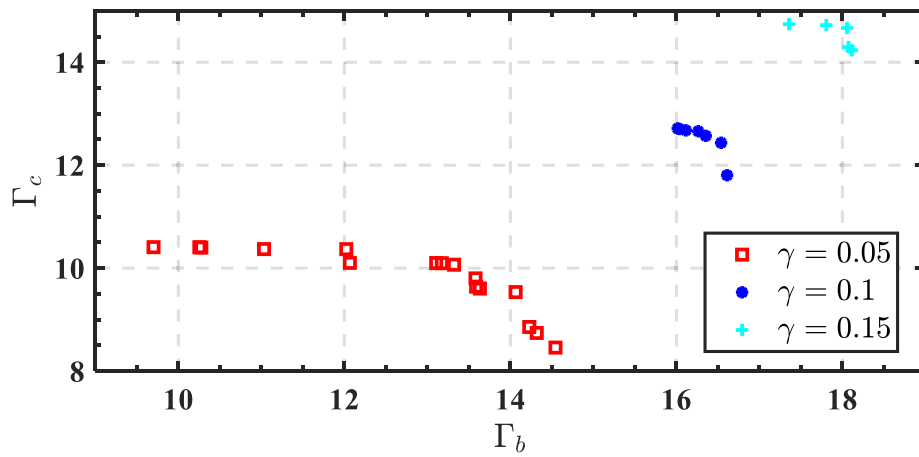
$$\begin{cases} \sigma_{ES,t}^{ch} + \sigma_{ES,t}^{dis} = 1 \\ 0 \leq P_{ES,t}^{ch} \leq \sigma_{ES,t}^{ch} P_{ES}^{ch-\max} \\ 0 \leq P_{ES,t}^{dis} \leq \sigma_{ES,t}^{dis} P_{ES}^{dis-\max} \end{cases} \quad (28)$$



(a) Wind-PV



(b) Wind-load



(c) PV-load

Fig. 9. Optimal Pareto front for wind-PV, wind-load and PV-load robustness objective, respectively.

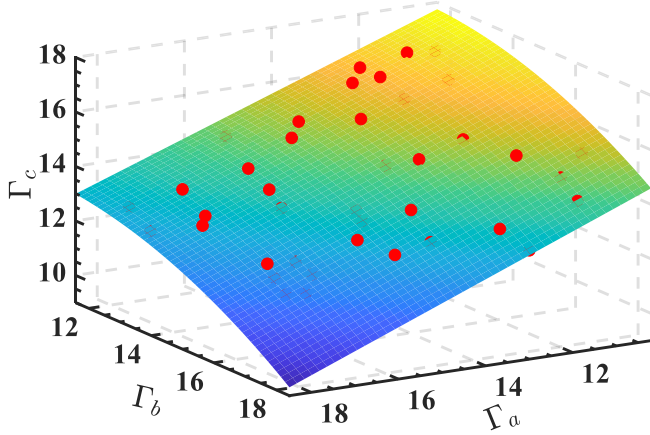


Fig. 10. Pareto front of robust optimal dispatching of IES based on MOCGDT when  $\lambda = 0.1$ .

Table 6

Pareto solutions for the four typical scenarios obtained by the MOCGDT when  $\lambda = 0.1$ .

| Scenario           | Robustness objective |            |            | Costs/CNY |
|--------------------|----------------------|------------|------------|-----------|
|                    | $\Gamma_a$           | $\Gamma_b$ | $\Gamma_c$ |           |
| Optimal Compromise | 14.52                | 16.75      | 14.02      | 10,207.58 |
| Optimal $\Gamma_a$ | 18.12                | 13.44      | 12.45      | 10,223.36 |
| Optimal $\Gamma_b$ | 12.76                | 18.10      | 12.78      | 10,100.89 |
| Optimal $\Gamma_c$ | 11.54                | 12.47      | 16.93      | 10,205.94 |

$$\begin{cases} \sigma_{HS,t}^{ch} + \sigma_{HS,t}^{dis} = 1 \\ 0 \leq H_{HS,t}^{ch} \leq \sigma_{HS,t}^{ch} H_{HS}^{ch-\max} \\ 0 \leq H_{HS,t}^{dis} \leq \sigma_{HS,t}^{dis} H_{HS}^{dis-\max} \end{cases} \quad (29)$$

where  $\sigma_{ES,t}^{ch}/\sigma_{HS,t}^{ch}$  and  $\sigma_{ES,t}^{dis}/\sigma_{HS,t}^{dis}$  denote the states of ES/HS during period  $t$ , respectively, and are represented as binary variables with a value of either 0 or 1.  $P_{ES}^{ch-\max}/H_{HS}^{ch-\max}$  and  $P_{ES}^{dis-\max}/H_{HS}^{dis-\max}$  are the upper limits of charging and discharging power of ES/HS, respectively.

Apart from the aforementioned constraints, there are ramp-up/down rate limits that apply to the operation of various energy conversion devices, including CHP, EB, EC, and AC, as specified in Eq. (30).

$$\begin{cases} P_{n,t} - P_{n-1,t} \leq UR_n \\ P_{n-1,t} - P_{n,t} \leq DR_n \end{cases} \quad (30)$$

where  $UR_n$  and  $DR_n$  represent the ramp-up and ramp-down rate limits, respectively, for device  $n$  in kw.

### 3.2. MOCGDT-based robust model

In this study, considering that the fluctuations of REG in different scheduling periods cannot be the same. According to the MOCGDT theory, the following bi-level robust optimal scheduling model can be obtained by integrating the wind-PV-load uncertain confidence interval Eqs. (5)-(7), optimization objective Eq. (10), and constraint Eqs. (11)-(30).

$$\begin{cases} \max_X \{ \Gamma_a, \Gamma_b, \Gamma_c \} \\ \left\{ \begin{aligned} \Gamma_a &= \sum_{t=1}^T (1 - \alpha_{W,t}), \Gamma_b = \sum_{t=1}^T (1 - \alpha_{S,t}), \Gamma_c = \sum_{t=1}^T (1 - \alpha_{L,t}) \\ 1 - \alpha_{W,t} &= \Pr_W \{ \Delta P_{W,t}^l \leq \Delta P_{W,t} \leq \Delta P_{W,t}^u \} \\ 1 - \alpha_{S,t} &= \Pr_S \{ \Delta P_{S,t}^l \leq \Delta P_{S,t} \leq \Delta P_{S,t}^u \} \\ 1 - \alpha_{L,t} &= \Pr_L \{ \Delta P_{L,t}^l \leq \Delta P_{L,t} \leq \Delta P_{L,t}^u \} \\ 0 \leq \alpha_{W,t} \leq 1, 0 \leq \alpha_{S,t} \leq 1, 0 \leq \alpha_{L,t} \leq 1 \\ f(X, P_{W,t}, P_{S,t}, P_{L,t}) &\leq f_c, f_c = (1 + \lambda)f_0 \\ \text{Eqs. (5) - (7), (11) - (30)} \end{aligned} \right. \end{cases} \quad (31)$$

where,  $f_0$  is the expected objective value obtained under the deterministic IES scheduling model.  $\Gamma_a$ ,  $\Gamma_b$  and  $\Gamma_c$  represent the robustness targets of wind power, PV and load, respectively.  $X$  represents the scheduling scheme.

The MOCGDT model cannot be directly solved and is dependent on the uncertainty variable fluctuations. This is essentially a bi-level optimization model with a multi-objective upper model to maximize the fluctuation range of wind-PV-load uncertainties while considering a predetermined operating cost, and a mixed-integer lower model to solve for the operating cost given a range of fluctuations and confirm it is less than the predetermined cost. As per the Eq. (10), the operating cost of IES increases when the wind-PV output is low, and the load demand is high. Therefore, the lower optimization model checks whether the operating cost meets the predetermined cost when taking the lower limit of wind-PV output and the upper limit of load, given a known confidence level.

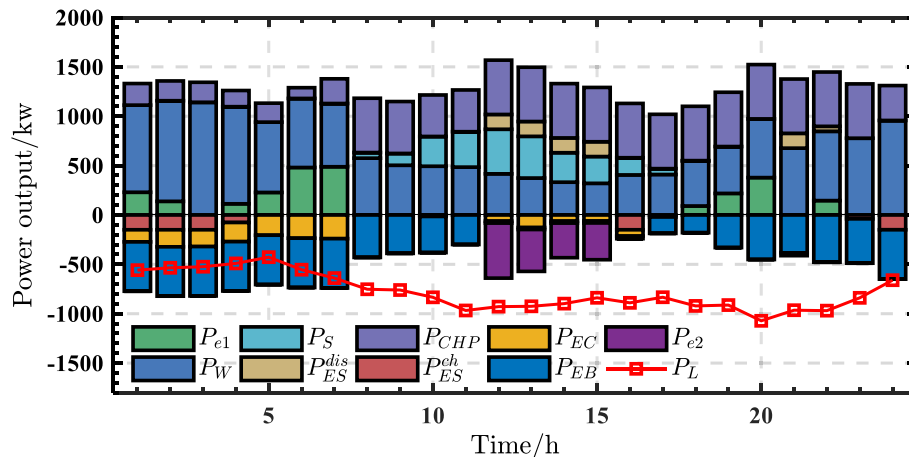


Fig. 11. Optimal dispatching for power supply.

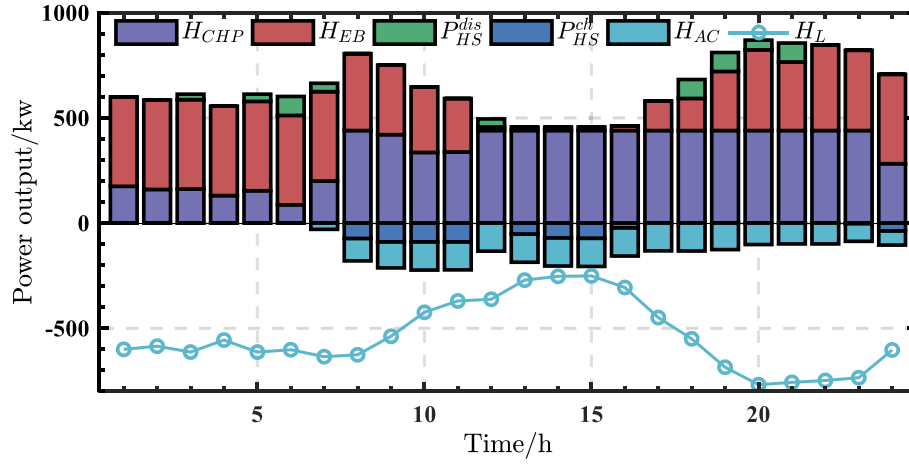


Fig. 12. Optimal dispatching for heat supply.

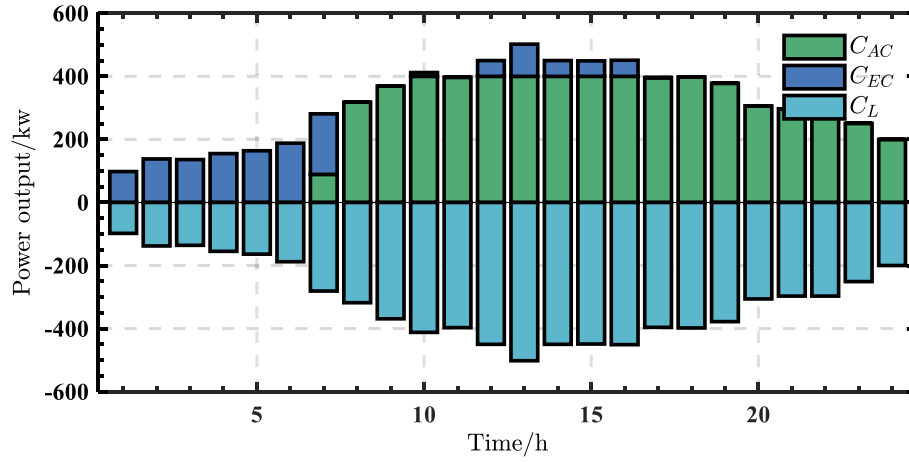


Fig. 13. Optimal dispatching for cool supply.

### 3.3. Solving algorithm of MOCGDT robust model

According to the characteristics of the MOCGDT model, an efficient bi-level solution algorithm is designed. Swarm-based intelligent algorithms have demonstrated natural advantages in solving such problems. The state transition algorithm (STA) (Zhou et al., 2012) is a popular and efficient optimization algorithm known for its fast convergence, robustness, and strong search capabilities. In recent years, it has gained widespread recognition and has been successfully implemented in various domains. Therefore, a population-based multi-objective STA (PMOSTA) (Dong et al., 2021) is used to solve the upper-level model. PMOSTA employs state transformation operators (STOs) to generate candidate populations, and then the next generation population is selected by combining the concepts of Pareto domination and crowding ranking in NSGA-II (Deb et al., 2002). The four STOs used in PMOSTA are as follows:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \varepsilon_\alpha \frac{1}{n \|\mathbf{x}_k\|_2} \mathbf{R}_r \mathbf{x}_k \quad (32)$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \varepsilon_\beta \mathbf{R}_l \frac{\mathbf{x}_k - \mathbf{x}_{k-1}}{\|\mathbf{x}_k - \mathbf{x}_{k-1}\|_2} \quad (33)$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \varepsilon_\gamma \mathbf{R}_c \mathbf{x}_k \quad (34)$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \varepsilon_\delta \mathbf{R}_a \mathbf{x}_k \quad (35)$$

where  $\varepsilon_\alpha$ ,  $\varepsilon_\beta$ ,  $\varepsilon_\gamma$  and  $\varepsilon_\delta$  are positive constants, called state transformation

factors.  $\mathbf{R}_r \in \mathbb{R}^{n \times n}$  is a matrix of randomly generated variables uniformly distributed in the interval  $[-1,1]$ .  $\mathbf{R}_l \in \mathbb{R}$  is a uniformly distributed random variable in the interval  $[0,1]$ .  $\mathbf{R}_e \in \mathbb{R}^{n \times n}$  is a diagonal matrix with randomly generated elements that follow a standard normal distribution.  $\mathbf{R}_a \in \mathbb{R}^{n \times n}$  is a diagonal matrix with randomly distributed elements following a standard normal distribution, except for a single position where a non-zero value is randomly assigned. The STOs in STA have different functions that satisfy the functional needs of global search, local search, and heuristic search, which enable STA to probabilistically find the global optimal Pareto front. For example, the candidate solutions generated by the rotation operator are all within a hypersphere of radius  $\varepsilon_\alpha$ , and the operator has global search capability when taking a large value; conversely, it has local exploitation capability. In addition, this paper proposes a non-linear control parameter strategy, presented in Eq. (36), to balance the global and local search ability of the rotation operator.

$$\varepsilon_\alpha = C_1 - (C_1 - C_2) \times \sin\left(\frac{\pi}{2} \left(\frac{g}{G_{\max}}\right)^2\right) \quad (36)$$

where  $g$  and  $G_{\max}$  denote the current iteration number and the maximum iteration number, respectively.  $C_1$  and  $C_2$  denote the upper limit and lower limit of the rotation factor, respectively;

For the solution of the lower-level model, when the confidence level of the upper-level is given, the range of uncertainties can be obtained, thus obtaining the extreme scenarios of wind-PV-load, as shown in Eq. (37).

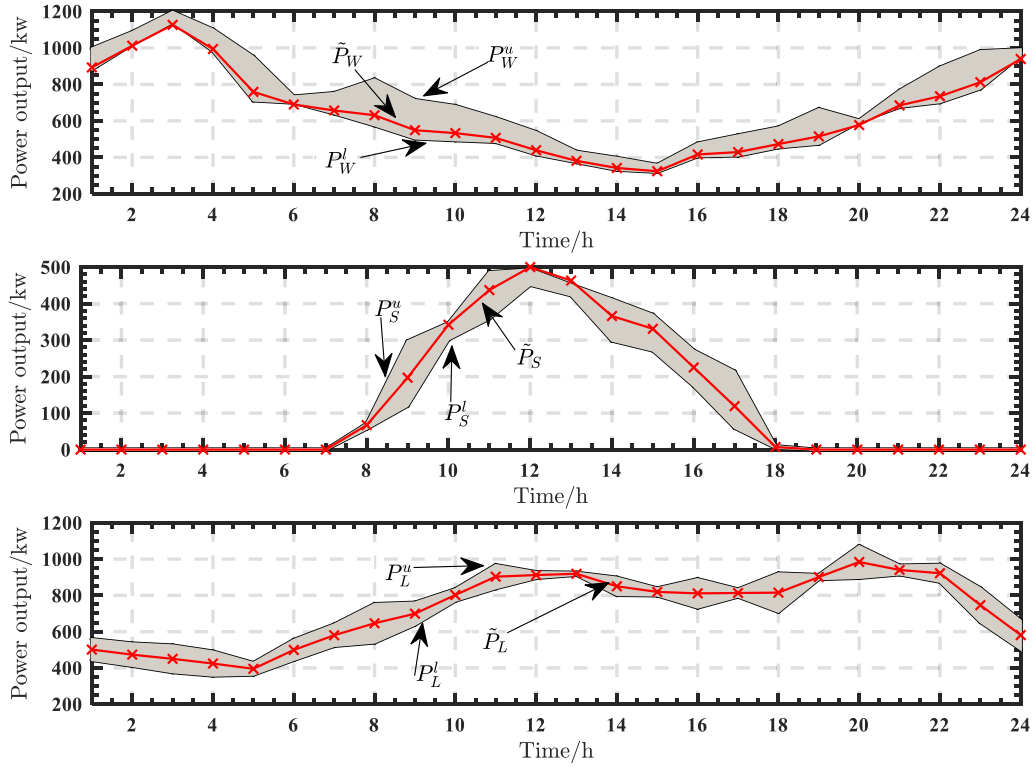


Fig. 14. Fluctuation interval of actual wind power, photovoltaic and load forecasts, respectively.

Table 7

The optimal compromise solutions of IES robust scheduling under different  $\lambda$ .

| $\lambda$ | Preset worst cost | Robustness objective |            |            | Costs/CNY |
|-----------|-------------------|----------------------|------------|------------|-----------|
|           |                   | $\Gamma_a$           | $\Gamma_b$ | $\Gamma_c$ |           |
| 0.05      | 9,766.64          | 11.48                | 13.81      | 8.47       | 9,753.90  |
| 0.1       | 10,231.72         | 14.52                | 16.75      | 14.02      | 10,207.58 |
| 0.15      | 10,696.80         | 19.82                | 17.34      | 16.72      | 10,668.83 |
| 0.2       | 11,161.87         | 19.87                | 19.82      | 19.65      | 11,145.51 |
| 0.25      | 11,626.95         | 19.91                | 20.59      | 21.07      | 11,618.54 |
| 0.3       | 12,092.03         | 21.83                | 22.08      | 21.89      | 12,012.96 |
| 0.35      | 12,557.12         | 22.84                | 22.96      | 22.82      | 12,460.42 |
| 0.4       | 13,022.19         | 23.65                | 23.72      | 23.32      | 12,995.36 |

Table 8

Comparison of the solution results of the proposed MOCGDT with RO and SO.

| Methods            | RO        | SO        | MOCGDT    |
|--------------------|-----------|-----------|-----------|
| Costs/CNY          | 12,871.92 | 10,618.79 | 10,207.58 |
| Computation time/s | 8.56      | 1,524.54  | 923.74    |

$$\begin{cases} P_{W,t} = \tilde{P}_{W,t} + F_{W,t}^{-1}\left(\frac{\alpha_{W,t}}{2}\right) \\ P_{S,t} = \tilde{P}_{S,t} + F_{S,t}^{-1}\left(\frac{\alpha_{S,t}}{2}\right) \\ P_{L,t} = \tilde{P}_{L,t} + F_{L,t}^{-1}\left(1 - \frac{\alpha_{L,t}}{2}\right) \end{cases} \quad (37)$$

It is not difficult to understand that the lower-level model is a mixed integer programming problem. In this paper, the YALMIP toolbox in the MATLAB platform is used for simulation, and CPLEX solver is called to solve it. Fig. 5 depicts the framework diagram of the proposed method for solving IES robust optimal scheduling.

## 4. Case study

### 4.1. Simulation setup

To validate the proposed MOCGDT model, this paper employs the IES illustrated in Fig. 4 for simulation. The parameters and data used in IES scheduling case are all from reference (Peng et al., 2021; Shi et al., 2019; L. Yang et al., 2018). The predicted values of wind power, PV, and loads are shown in Fig. 6; the natural gas price is 2.5 CNY/m<sup>3</sup>; The parameters of IES and the electricity prices are shown in Table 4 and Table 5, respectively; the parameters of PMOSTA algorithm are set as follows: the population size is 40, the maximum number of iterations is 40,  $\varepsilon_\beta = \varepsilon_\gamma = \varepsilon_\delta = 1$ ,  $C_1 = 1$  and  $C_2 = 1e-2$ .

### 4.2. Simulation analysis

#### 4.2.1. Robust scheduling with single uncertainty

In this subsection, only the effect of a single uncertain variable of wind-PV-load on the robust scheduling results is considered, and the upper model is a single-objective problem, which can be solved by using the basic STA algorithm. Firstly, we get the optimal operating cost 9301.56 CNY by solving the deterministic model, and then according to different expected operating costs of decision-makers, curves of economic cost and robustness varying with the objective deviation factor can be obtained, as depicted in Fig. 7 and Fig. 8.

Fig. 7 and Fig. 8 reveal a positive correlation between the objective deviation factor and both operating cost and robustness. In the robustness model, the maximum fluctuation confidence level of wind-PV-load, both increase with the increase of the objective deviation factor, that is, the larger the acceptable maximum fluctuation range of wind-PV-load when the expected operation cost is high, the better the robustness of the obtained dispatching strategy is and the more it can resist large uncertain fluctuations. It should be emphasized that when the objective deviation factor reaches a certain threshold, the maximum confidence level of wind-PV-load for each dispatching period will reach the



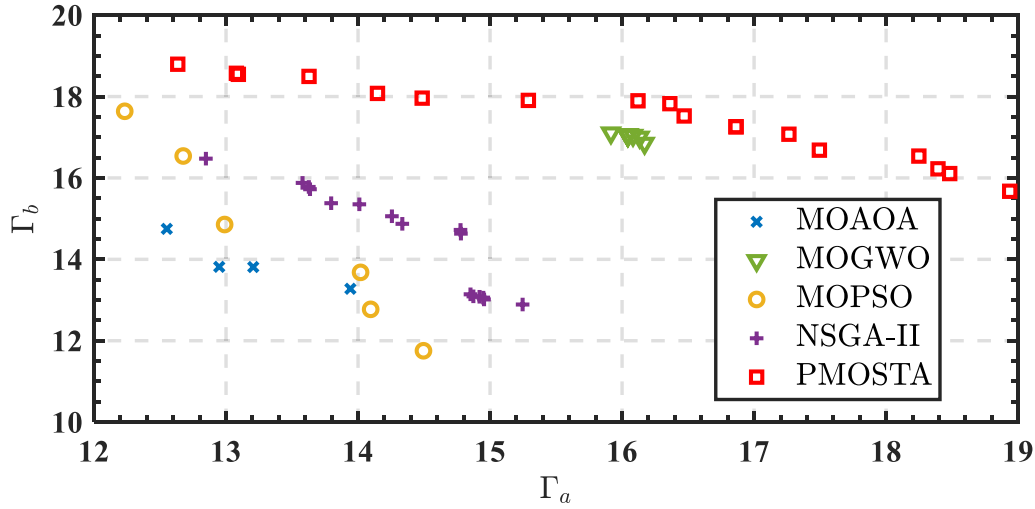


Fig. 15. Comparison of Pareto fronts obtained by five multi-objective algorithms.

Table 9

Comparison of compromise solutions obtained by five multi-objective algorithms.

| Algorithms | MOAOA | MOGWO | MOPSO | NSGA-II | PMOSTA |
|------------|-------|-------|-------|---------|--------|
| $\Gamma_a$ | 13.21 | 16.08 | 14.02 | 14.77   | 16.36  |
| $\Gamma_b$ | 13.81 | 17.05 | 13.68 | 14.72   | 17.82  |

maximum value, i.e.  $\Gamma_a = \Gamma_b = \Gamma_c = 24$ ,  $1 - \alpha_{W,t} = 1 - \alpha_{S,t} = 1 - \alpha_{L,t} = 1$ . Specifically, when the wind-PV-load uncertainty reaches the maximum robustness, its corresponding objective deviation factors are 0.35, 0.15, and 0.2, respectively. Obviously, robust scheduling considering single uncertainty has certain practical engineering value and can provide a theoretical reference for schedulers, but it cannot coordinate optimization among multiple uncertainties and has certain limitations.

#### 4.2.2. Robust scheduling considering multiple uncertainties simultaneously

To validate the effectiveness of the MOCGDT robust model, this study considers multiple uncertain variables and applies the model to IES robust scheduling. The model is solved using the developed bi-level optimization algorithm. Fig. 9 shows the optimal Pareto fronts obtained by the PMOSTA algorithm when two uncertainty variables are considered simultaneously under different objective deviation factors. It can be observed from Fig. 9 that any two robust objectives under the same objective factor are in conflict with each other, and the overall robustness of the Pareto solution set obtained by the PMOSTA becomes stronger as the objective deviation factor increases, reflecting that the scheduling scheme becomes more tolerant to the deteriorating operating cost.

Fig. 10 depicts the Pareto front surface obtained through PMOSTA algorithm for the wind-PV-load uncertainty when setting the objective deviation factor to 0.1. Table 6 lists four typical solutions in the Pareto solution set including optimal  $\Gamma_a$ , optimal  $\Gamma_b$ , optimal  $\Gamma_c$ , and optimal compromise solution, where the compromise solution is obtained according to fuzzy decision theory (Peng et al., 2019). Table 6 shows that the optimal operating costs in the four scenarios are quite similar, but the robust objectives vary significantly. Specifically, since there is conflicting nature among the three objectives, the Pareto optimal solution obtained under considering only one of the objectives optimally is relatively inferior to the other two objectives. By contrast, although none of the objectives of the compromise solution is optimal, each objective is acceptable on the whole, which would be a relatively ideal solution for the decision-maker.

Figs. 11–13 show the scheduling results corresponding to the

compromise solution for  $\lambda = 0.1$ , respectively. Figs. 11–13 reveal that during high wind power output periods with low electric and cooling load demand but high heating load, CHP unit output decreases and electric boiler is mainly utilized for heating, while electric chiller can meet cooling load demand. Electric energy storage is in charging state. During the day with high electrical load but low heating load, excess heat energy is stored in the heat storage. Electric storage is in discharge state, CHP unit output increases, and electricity is sold to the grid during peak electricity prices to reduce operating costs effectively.

Fig. 14 shows the fluctuation interval of wind-PV-load corresponding to the compromise solution in each scheduling period when  $\lambda = 0.1$ , which is not symmetric and can better reflect the actual uncertain fluctuation situation. Table 7 lists the optimal compromise solutions obtained for IES optimal scheduling based on the MOCGDT robust model for different objective deviation factors. Table 7 shows that as the objective deviation factor increases, the acceptable operating cost increases, while the robustness objective obtained is enhanced. This result is consistent with the previous conclusions obtained from the robust scheduling experiments considering only one type of uncertainty.

#### 4.2.3. Comparisons with RO and SO

To validate the superiority of the proposed MOCGDT robust model, we compare it with two popular uncertain optimization methods, RO and SO. When the objective deviation factor of MOCGDT is set to 0.1, Table 8 shows the optimization results of the three methods for the IES scheduling problem with wind-PV-load multiple uncertainties. From Table 8, it can be seen that MOCGDT, RO, and SO obtained scheduling costs of 10,207.58 CNY, 12,871.92 CNY, and 10,618.79 CNY, respectively. MOCGDT achieved cost improvements of 20.69% and 3.87% compared to RO and SO, respectively. In addition, the solution time of RO is the shortest, and the solution time of MOCGDT is 923.74 s, which is 601 s less than that of SO. This is because MOCGDT belongs to a bi-level model, where the upper level uses a stochastic optimization algorithm, which requires a longer iterative calculation. On the other hand, SO needs to construct more scenarios, leading to a higher computational complexity. In summary, the proposed MOCGDT robust model obtains less conservative results compared to RO, and has lower computational complexity than SO. Therefore, it has significant advantages over both RO and SO.

#### 4.2.4. Performance comparison of multi-objective optimization algorithms

To demonstrate the effectiveness of the PMOSTA algorithm, we compared it with MOAOA (Khodadadi et al., 2022), MOGWO (Mirjalili et al., 2016), NSGA-II (Deb et al., 2002) and MOPSO (Mostaghim & Teich, 2003), by applying them to solve the IES optimal scheduling

problem. In the experiment, the population of all five algorithms is 40 and the maximum number of iterations is 40. Taking the robust scheduling considering both wind power and PV uncertainties as an example and setting the objective deviation factor to 0.05, Fig. 15 shows the Pareto frontier obtained by the five algorithms. Owing to the powerful search capability of the state transformation operator of PMOSTA and the nonlinear parameter control strategy, the global exploration and local exploitation capability of the algorithms are effectively balanced, and a more accurate and diversified optimization search process can be achieved. Fig. 15 shows that the Pareto front obtained by PMOSTA has the widest span and is located above the Pareto fronts obtained by MOAOA, MOGWO, NSGA-II and MOPSO. This indicates that the PMOSTA achieves better accuracy and diversity of solutions.

Table 9 lists the optimal compromise solutions obtained by five multi-objective optimization algorithms. From Table 9, it can be seen that under the preset objective cost, the values of  $\Gamma_a$  and  $\Gamma_b$  obtained by the PMOSTA algorithm are greater than those obtained by the other four algorithms, indicating that PMOSTA achieves the best robustness, and that the optimization performance of PMOSTA is significantly superior to MOAOA, MOGWO, NSGA-II, and MOPSO.

## 5. Conclusion

This paper proposes a MOCGDT robust optimal scheduling model for IES to address multiple uncertainties. The model aims to maximize the fluctuation range of each uncertain variable in the upper level and minimize the operating cost in the lower level. An efficient bi-level solution algorithm is developed to solve the model. To overcome the limitation of the coarse box set in describing uncertain variables, the proposed robust model uses a GMM to construct the confidence fluctuation interval of the REG forecasting error. The model also considers the variability of fluctuation magnitude of the same uncertain variable in different scheduling periods and the conflict between fluctuations of different uncertain variables when the budget cost is fixed. The simulation results show that the scheduling cost of the proposed MOCGDT model is 10,207.58 CNY, which is improved by 20.69% and 3.87% compared with RO and SO, respectively. The conservatism of MOCGDT is reduced and more reasonable. In addition, the computation time of MOCGDT is reduced by 601 s compared with SO. Therefore, the proposed model has obvious advantages over RO and SO. Moreover, the Pareto frontier obtained by PMOSTA algorithm is better than that obtained by MOAOA, MOGWO, NSGA-II and MOPSO, indicating that PMOSTA has the best optimization performance.

It should be noted that the proposed MOCGDT model in this paper is not without limitations. It can be observed that the lower-level model is restricted to a single-objective optimization problem. Specifically, when considering multiple optimization objectives at the lower level, planners may not wish to convert the multi-objective optimization problem into a linear weighting problem, but rather seek a Pareto solution set, which introduces difficulties in decision-making for the lower-level objectives and significantly increases the computational complexity of the model. Therefore, in future work, we plan to improve the MOCGDT model and apply it to IES planning, multi-regional IES collaborative operation, energy storage capacity configuration, and other fields.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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